FOR COMPLETE BEGINNERS

A cart that adds itself up.

Complete Modern Java, episode 20: a list of objects, in Java and TypeScript

> Move the list and the loop into the class that owns them, and let main just ask.

A list of names cannot add up prices. A list of Product objects can.

A ShoppingCart owns the list and works out its own total; main only asks.

In this episode

- ArrayList<Product>
- A class that owns a list
- import in a class file
- add, count, totalCents
- Refusing null
- TypeScript cart

Coming next

- Removing products, and what makes two products the same (episode 21)

At the time of recording: Java 25, Node 24 LTS, IntelliJ IDEA 2026.1.3.

01 What each run printed

Groceries.java: the loop in main	`499`
Shopping.java: empty cart, then three products	`0, 3, 499`
cart.add(null), no check	`count 4, then NullPointerException in totalCents`
cart.add(null), with the check	`IllegalArgumentException: product cannot be null`
shoppingCart.ts	`2, 249`
shoppingCart.ts: cart.add(null)	`tsc TS2345: null is not assignable to Product`

Every row was run, not predicted.

02 The class that owns the list

A normal class file needs import java.util.ArrayList; the compact files got it without one. Product, our own class in the same folder, needs no import.

The cart checks what it accepts: private alone does not stop a null; add does.

ShoppingCart.java

```
import java.util.ArrayList;

class ShoppingCart {
    private ArrayList<Product> items = new ArrayList<>();

    void add(Product product) {
        if (product == null) {
            throw new IllegalArgumentException("product cannot be null");
        }
        items.add(product);
    }

    int count() { return items.size(); }

    int totalCents() {
        int total = 0;
        for (Product product : items) {
            total += product.getPriceCents();
        }
        return total;
    }
}
```

> Shopping.java printed 0, 3, 499; with the check, cart.add(null) threw IllegalArgumentException.
>
> Run on a Mac, 2026-11-07: Java 25.0.4, Node 24.22.0, TypeScript 7.0.2.

03 main just asks

Before, main did all the work. Now main tells the cart what it wants, and the cart knows how to do it: one place to change the total.

Shopping.java

```
void main() {
    ShoppingCart cart = new ShoppingCart();
    IO.println(cart.totalCents());   // 0
    cart.add(new Product("Bread", 150));
    cart.add(new Product("Milk", 99));
    cart.add(new Product("Eggs", 250));
    IO.println(cart.count());        // 3
    IO.println(cart.totalCents());   // 499
}
```

> Run on Java 25.0.4 with Product.java from Episode 17.
>
> Run on a Mac, 2026-11-07: Java 25.0.4, Node 24.22.0, TypeScript 7.0.2.

04 TypeScript: the checker refuses null

The same cart with a Product[] and push. Passing null to add is a tsc error before the code runs.

shoppingCart.ts

```
class ShoppingCart {
    private items: Product[] = [];
    add(product: Product): void { this.items.push(product); }
    count(): number { return this.items.length; }
    totalCents(): number {
        let total = 0;
        for (const product of this.items) { total += product.priceCents; }
        return total;
    }
}
```

> node printed 2 and 249; cart.add(null) was tsc error TS2345.
>
> Run on a Mac, 2026-11-07: Java 25.0.4, Node 24.22.0, TypeScript 7.0.2.

Give data a home

1. The class that owns the data does the work.
2. Keep the list private; offer methods.
3. Check what add accepts.
4. A class file imports ArrayList.
5. Challenge: add Rice at 199. Count and total?

Everything else

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The code and the challenge files: complete-modern-java
https://github.com/code-with-sam-dev/complete-modern-java

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